

Optimal lattices for a three-body power-law energy: steep-decay limits, computer-assisted proofs and the minima of the modular graph functions $C_{s,s,s}$

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Abstract We study which Bravais lattice of fixed density minimises the three-body energy $T_v(\Lambda) = \sum'_{x,y \in \Lambda} (|x||y||x-y|)^{-v}$, the sum over all lattice triangles through the origin of the product of their side lengths to the power $-v$. In two dimensions T_{2s} coincides, up to a factor, with the two-loop modular graph function $C_{s,s,s}(\tau)$. We prove that in the steep-decay limit $v \rightarrow \infty$ the minimisers converge to the rectangular lattice with aspect ratio $\sqrt{(\sqrt{17}-1)/2}$ in $d=2$, by solving exactly the associated max–min problem for the smallest triangle product, and to the body-centred cubic lattice in $d=3$, by a computer-assisted argument. For finite exponents we give computer-assisted proofs that $C_{2,2,2}$, $C_{3,3,3}$ and $C_{4,4,4}$ attain their minimum only at the square point $\tau = i$ (whereas $C_{1,1,1}$ is minimal at the hexagonal point), that the square lattice is the unique minimiser also for $v=5,7$, that the hexagonal lattice is the unique minimiser for $v=3.5$, and that the hexagonal and square energies cross in $(3.918364, 3.918366)$. The proofs rest on a majorant series that bounds all Taylor coefficients of every summand uniformly in the lattice vectors. Numerically, the planar minimiser passes from hexagonal to square at $v^* \approx 3.91836$ (first order) and from square to rectangular at $v_2 = 8.6063 \pm 0.0005$; in three dimensions BCC is optimal for all tested exponents, FCC loses local stability at $v_c = 3.7521 \pm 0.0001$, and hcp is never optimal.

Keywords Lattice energy minimisation · Three-body interactions · Modular graph functions · Computer-assisted proof · Max–min problems · Crystallisation

1 Introduction

Which periodic arrangement of points minimises a given interaction energy is one of the classical questions linking number theory, geometry and the physics of crystals. For *pair* interactions a great deal is known: for completely monotone functions of the squared distance the hexagonal lattice is optimal in the plane among all lattices of fixed density [1, 2, 3, 4, 5], universal optimality has been established for E_8 and the Leech lattice [6], and phase transitions between optimal lattices have been classified for many two-term and ratio-type pair energies [7, 8, 9]. In three dimensions the optimality of the face-centred cubic lattice for Epstein zeta functions (the Sarnak–Strömbergsson conjecture) has recently been claimed [10].

Much less is known for genuine *many-body* energies. The

simplest one is the product-form three-body power law

$$T_v(\Lambda) = \sum_{\substack{x,y \in \Lambda \setminus \{0\} \\ x \neq y}} (|x||y||x-y|)^{-v}, \quad (1)$$

which converges for $v > 2d/3$. At $v=3$ it is the radial part of the Axilrod–Teller–Muto (ATM) triple-dipole energy; under the name *three-body zeta function* it was introduced and evaluated for selected lattices and along the Bain path in [11], and efficient algorithms for such many-body lattice sums were developed in [12, 13]. We stress that (1) is a model energy: physical three-body dispersion carries an angular factor and always comes with pair terms. Our motivation is mathematical. In $d=2$, the energy T_{2s} is (up to normalisation) the two-loop *modular graph function* $C_{s,s,s}(\tau)$ from the low-energy expansion of genus-one string amplitudes [15]; its minimum over the fundamental domain appears not to have been studied.

Two remarks delimit the content. First, T_v is invariant under isometries, so the hexagonal and the square lattice (the elliptic fixed points $\rho = e^{i\pi/3}$ and i of the modular group) are always critical points [16]. The substance of the results below is *global* minimality and the location of transitions. Second, the reduction of a steep-decay limit to a packing-type problem is standard for pair energies [7]; here it leads to a new max–min problem for triangles, related in spirit to three-point packing problems on the sphere [17].

Summary of results. Fig. 1 summarises the planar picture.

- Theorem 1 solves exactly the planar max–min problem for the smallest triangle product; Theorem 2 deduces that minimisers of T_v converge to a rectangular lattice with aspect ratio $y_\infty = \sqrt{(\sqrt{17}-1)/2}$.
- Theorem 3 (computer-assisted) shows that in $d=3$ the body-centred cubic (BCC) lattice is the unique maximiser of the smallest triangle product; minimisers of T_v converge to BCC.
- Theorem 4 (computer-assisted) establishes the minimiser for the finite exponents $v \in \{3.5, 4, 5, 6, 7, 8\}$; in particular $C_{2,2,2}$, $C_{3,3,3}$, $C_{4,4,4}$ are minimal only at $\tau = i$.
- Sections 6 and 7 give the numerical phase diagrams with error bars, including hcp, unequal exponents and mixed pair plus three-body energies.

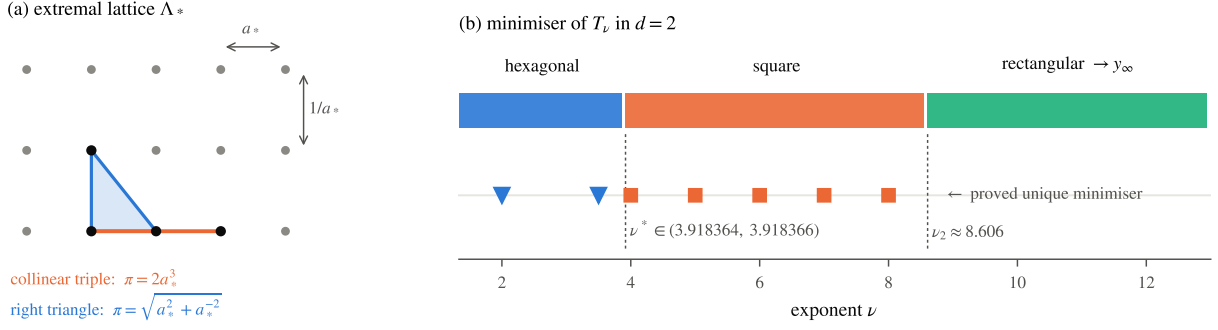


Fig. 1 (a) The extremal lattice Λ_* of Theorem 1: the collinear triple (red) and the right triangle (blue) have the same side-length product P_* . (b) Minimiser of T_ν in $d=2$. Shaded regions: numerical phase diagram (Sect. 6); symbols: exponents at which the minimiser is established rigorously ($\nu=2$ classical, the others by Theorem 4). For $\nu \rightarrow \infty$ the minimiser converges to Λ_* (Theorem 2).

2 Setting and the modular graph function

Throughout, lattices are Bravais lattices of unit covolume, considered up to isometry. For three pairwise distinct points p, q, r put $\pi(p, q, r) = |p - q||q - r||r - p|$, and let

$$P(\Lambda) = \min\{\pi(p, q, r) : p, q, r \in \Lambda \text{ pairwise distinct}\},$$

the smallest triangle product; collinear triples are admitted. Every triple can be translated to contain the origin, hence

$$T_\nu(\Lambda) \geq P(\Lambda)^{-\nu}. \quad (2)$$

In $d=2$ we write $\Lambda_\tau = (\mathbb{Z} + \tau\mathbb{Z})/\sqrt{\text{Im } \tau}$ with $\tau = x + iy$ in the fundamental domain. The modular graph function of [15] is $C_{a,b,c}(\tau) = \sum_{p_1+p_2+p_3=0, p_i \neq 0} \prod_i (\text{Im } \tau / \pi |p_i|^2)^{a_i}$, $p_i \in \mathbb{Z} + \tau\mathbb{Z}$, and a direct comparison gives

$$C_{s,s,s}(\tau) = \pi^{-3s} T_{2s}(\Lambda_\tau). \quad (3)$$

At $s=1$, Zagier's identity $C_{1,1,1} = E_3 + \zeta(3)$ and the minimality of the Epstein zeta function at ρ [1, 2, 3, 4] show that $C_{1,1,1}$ is minimal only at ρ . We verified (3) numerically, and the identities $C_{1,1,1} = E_3 + \zeta(3)$ and $C_{2,2,1} = \frac{2}{3}E_5 + \zeta(5)/30$ to relative accuracy 10^{-14} and 10^{-15} at five values of τ (Appendix A).

3 The planar steep-decay limit

Let $a_* = ((1 + \sqrt{17})/8)^{1/4} \approx 0.894563$, so that $4a_*^8 = a_*^4 + 1$, and put

$$P_* = 2a_*^3 = \sqrt{a_*^2 + a_*^{-2}} \approx 1.431734, \quad \Lambda_* = a_*\mathbb{Z} \times a_*^{-1}\mathbb{Z}.$$

The aspect ratio of Λ_* is $y_\infty = a_*^{-2} = \sqrt{(\sqrt{17} - 1)/2} \approx 1.249621$.

Theorem 1. *For every lattice $\Lambda \subset \mathbb{R}^2$ of unit covolume, $P(\Lambda) \leq P_*$, with equality if and only if Λ is isometric to Λ_* .*

Lemma 1. *After a rotation and a reflection, Λ has a basis $u = (a, 0)$, $v = (p, a^{-1})$ with $a = \lambda_1(\Lambda)$, $0 \leq p \leq a/2$ and $|v| \geq a$; moreover $a^4 \leq 4/3$.*

Proof. Lagrange–Gauss reduction; the bound is Hermite's constant $\gamma_2 = 2/\sqrt{3}$. $\square$

Lemma 2. *In the notation of Lemma 1, $P(\Lambda) \leq G(a) := \min\{2a^3, \sqrt{a^2 + a^{-2}}\}$, and $\pi(0, u, v) \leq \sqrt{a^2 + a^{-2}}$ with equality if and only if $p=0$.*

Proof. The collinear triple $(0, u, 2u)$ gives $2a^3$. For $(0, u, v)$ we have $\pi^2 = a^2(p^2 + k)((a - p)^2 + k)$ with $k = a^{-2}$. With $m = p(a - p) \in [0, a^2/4]$ and $p^2 + (a - p)^2 = a^2 - 2m$,

$$(p^2 + k)((a - p)^2 + k) = (ka^2 + k^2) + m(m - 2k).$$

Since $a^4 < 8$, $m \leq a^2/4 < 2k$, so the last term is ≤ 0 , with equality iff $m=0$, i.e. $p=0$. Hence $\pi^2 \leq a^2(ka^2 + k^2) = a^2 + a^{-2}$. $\square$

Lemma 3. *For $0 < a \leq (4/3)^{1/4}$, $G(a) \leq P_*$ with equality iff $a = a_*$.*

Proof. For $a \leq a_*$, $G(a) \leq 2a^3 \leq 2a_*^3$. The function $g(a) = a^2 + a^{-2}$ decreases on $(0, 1]$ and increases on $[1, \infty)$. For $a_* < a \leq 1$, $G(a) \leq \sqrt{g(a)} < \sqrt{g(a_*)} = P_*$. For $1 \leq a \leq (4/3)^{1/4}$, $g(a) \leq \frac{2}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{7}{2\sqrt{3}}$ and $(\frac{7}{2\sqrt{3}})^2 = \frac{49}{12} < 16(\frac{1+\sqrt{17}}{8})^3 = P_*^4$. $\square$

Lemma 4. $P(\Lambda_*) = P_*$.

Proof. The triples in Fig. 1(a) give $P(\Lambda_*) \leq P_*$. Collinear triples lie on a lattice line with spacing $s \geq a_*$ and have $\pi = s^3(k - j)(l - k)(l - j) \geq 2s^3 \geq P_*$. A non-collinear lattice triangle has area $\geq \frac{1}{2}$; with side lengths $d_1 \leq d_2 \leq d_3$ and γ the angle between the two shorter sides, $d_1 d_2 \geq d_1 d_2 \sin \gamma \geq 1$, hence $\pi \geq d_3$. The only vectors of Λ_* shorter than $D = \sqrt{a_*^2 + a_*^{-2}}$ are $\pm u, \pm v$ with $u = (a_*, 0)$, $v = (0, a_*^{-1})$: for (ma_*, na_*^{-1}) with $n \neq 0$ one gets $m=0$ and $n^2 < 1 + a_*^4 < 2$, and for $n=0$, $m^2 < 1 + y_\infty^2 < 4$. They span two directions, whereas the three sides of a triangle are pairwise linearly independent. Thus $d_3 \geq D = P_*$. $\square$

Proof of Theorem 1. Lemmas 1–3 give $P(\Lambda) \leq G(a) \leq P_*$. Equality forces $a = a_*$ and $\pi(0, u, v) \geq P_* = \sqrt{a_*^2 + a_*^{-2}}$, hence $p=0$ by Lemma 2, i.e. $\Lambda = \Lambda_*$. The converse is Lemma 4. $\square$

Theorem 2. *For every $\nu > 4/3$, T_ν attains its minimum among planar lattices of unit covolume. If Λ_ν is any minimiser, then $\Lambda_\nu \rightarrow \Lambda_*$ up to isometries as $\nu \rightarrow \infty$, and*

$(\min T_V)^{1/\nu} \rightarrow 1/P_*$. Equivalently, minimisers τ_s of $C_{s,s,s}$ on the fundamental domain satisfy $\tau_s \rightarrow iy_\infty$ as $s \rightarrow \infty$.

Proof. P is continuous and $P(\Lambda) \leq 2\lambda_1(\Lambda)^3$; T_V is lower semicontinuous as a sum of non-negative continuous functions. By (2), sublevel sets of T_V lie in $\{\lambda_1 \geq \varepsilon\}$, compact by Mahler's criterion, so minimisers exist. Every term of $T_V(\Lambda_*)$ has $\pi \geq P_*$, whence $T_V(\Lambda_*) \leq CP_*^{-\nu}$ for $\nu \geq \nu_0$ with $C = T_{V_0}(\Lambda_*)P_*^{\nu_0}$. With (2), $P(\Lambda_\nu)^{-\nu} \leq T_V(\Lambda_\nu) \leq T_V(\Lambda_*) \leq CP_*^{-\nu}$, so $P_* \geq P(\Lambda_\nu) \geq C^{-1/\nu}P_* \rightarrow P_*$. The Λ_ν lie in a compact set; every limit point has $P = P_*$ and is Λ_* by Theorem 1. The statement for $C_{s,s,s}$ follows from (3). $\square$

The numerical minimisers for finite ν (Sect. 6) approach Λ_* monotonically: the optimal aspect ratio is 1.056, 1.100, 1.141, 1.170, 1.194 and 1.215 at $\nu = 9, 10, 12, 15, 20, 30$. A random test on $2 \cdot 10^4$ lattices confirmed the bound of Lemma 2, and a grid maximisation of P located the maximum at $\tau \approx 1.2475i$.

4 Three dimensions

In $d = 3$ the energy converges for $\nu > 2$. For BCC of unit covolume $P(\text{BCC}) = 3/2$, attained by isosceles triangles with two nearest-neighbour sides and one second-neighbour side; in contrast $P(\text{FCC}) = P(\text{SC}) = P(\text{hcp}) = \sqrt{2}$.

Theorem 3 (computer-assisted). *For every lattice $\Lambda \subset \mathbb{R}^3$ of unit covolume, $P(\Lambda) \leq 3/2$, with equality iff Λ is isometric to BCC. Consequently, for $\nu > 2$ minimisers of T_V exist and converge to BCC as $\nu \rightarrow \infty$, with $(\min T_V)^{1/\nu} \rightarrow 2/3$.*

Let $G = B^T B$ be the Gram matrix of a basis, normalised by $G_{11} = \lambda_1^2 = 1$, and $Q(G) = P(\Lambda)/\sqrt{\det G}$ (scale invariant). The remaining entries $q = (g_{22}, g_{33}, g_{12}, g_{13}, g_{23})$ are the parameters.

Lemma 5. *Every lattice has, up to isometry and scaling, a Gram matrix in the region $\mathcal{R}$ given by $1 \leq g_{22} \leq g_{33}$, $0 \leq g_{12}, g_{13} \leq \frac{1}{2}$, $|g_{23}| \leq g_{22}/2$ and $1 + g_{22} + 2(\varepsilon_1 \varepsilon_2 g_{12} + \varepsilon_1 g_{13} + \varepsilon_2 g_{23}) \geq 0$ for $\varepsilon_i = \pm 1$. On $\mathcal{R}$, $\det G \geq g_{22}g_{33}/4$; hence $Q < 3/2$ whenever $g_{22}g_{33} > 64/9$.*

Proof. Take a Minkowski-reduced basis and flip the signs of b_2, b_3 . The triangle $(0, b_1, b_2)$ is acute, so the covering radius of $L_2 = \mathbb{Z}b_1 + \mathbb{Z}b_2$ is its circumradius ρ , with $\rho^2 = g_{22}|b_2 - b_1|^2/(4(g_{22} - g_{12}^2)) \leq (1 + g_{22})/3$. Minkowski reduction places the projection of b_3 onto the plane of L_2 in its Voronoi cell, so the height h of b_3 satisfies $h^2 \geq g_{33} - \rho^2 \geq g_{33}/3$, and $\det G = (g_{22} - g_{12}^2)h^2 \geq g_{22}g_{33}/4$. The collinear triple $(0, b_1, 2b_1)$ gives $Q \leq 2/\sqrt{\det G}$. $\square$

Within $\mathcal{R}$, BCC has the single representative $q_0 = (1, 1, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3})$, found by enumerating unimodular changes of basis.

Lemma 6 (local lemma). *For a triangle $t = (0, m, n)$ put $f_t(q) = \sqrt{\ell_m \ell_n \ell_{m-n}} / \det G$ with $\ell_v = v^T G v$. At q_0 exactly 36 triangles are active, $f_t(q_0) = 3/2$, and their gradients take 6 distinct values whose convex hull contains the origin at distance $c = 0.13887$ from its boundary. On $q_0 + [-r, r]^5$ with $r = 0.007$, interval arithmetic gives $\|\nabla^2 f_t\|_F \leq M = 14.92$ for all active t . Hence $\min_i f_t(q_0 + \delta) \leq \frac{3}{2} - c|\delta| + \frac{M}{2}|\delta|^2 < \frac{3}{2}$ for $0 < |\delta| \leq \sqrt{5}r = 0.0157 < 2c/M = 0.0186$.*

Branch and bound. On a box B of parameters each ℓ_v is linear in q and $\det G$ is a cubic polynomial, so $Q \leq \min_{t \in S} (\prod \max_B \ell)^{1/2} (\min_B \det G)^{-1/2}$ for any set S of candidate triangles; we use all $(0, m, n)$ with $m, n \in \{-1, 0, 1\}^3$ and the collinear $(0, m, 2m)$. Starting from the compact box $[1, \frac{8}{3}] \times [1, \frac{64}{9}] \times [0, \frac{1}{2}]^2 \times [-\frac{4}{3}, \frac{4}{3}]$, boxes are discarded when they violate a constraint of $\mathcal{R}$, lie in $q_0 + [-r, r]^5$, or have upper bound $< 3/2$, and bisected otherwise. The run terminates after $9.93 \cdot 10^6$ boxes with every box certified. With Lemmas 5 and 6 this proves $Q \leq 3/2$ on $\mathcal{R}$ with equality only at q_0 . The convergence statement follows as in Theorem 2.

5 Finite exponents: computer-assisted proofs

Theorem 4 (computer-assisted). *Among planar lattices of unit covolume, the square lattice is the unique minimiser of T_V for $\nu \in \{4, 5, 6, 7, 8\}$, and the hexagonal lattice is the unique minimiser for $\nu = 3.5$. In particular $C_{2,2,2}$, $C_{3,3,3}$ and $C_{4,4,4}$ attain their minimum on the fundamental domain only at $\tau = i$. Moreover $T_V(\mathbb{Z}^2) - T_V(\Lambda_\rho)$ changes sign in $(3.918364, 3.918366)$.*

Together with the case $s = 1$ and Theorem 2, the minimiser of $C_{s,s,s}$ is thus ρ for $s = 1$, i for $s = 2, 3, 4$, and tends to iy_∞ as $s \rightarrow \infty$.

A universal majorant. For $\nu = (m, n)$ and $\tau = x + iy$ let $\ell_v = |m + n\tau|^2/y$, so that $T_V = \sum \prod_{\text{sides}} \ell^{-s}$ with $s = \nu/2$. Around (x_0, y_0) put $A = a/y_0$, $B = b/y_0$. A short computation gives

$$\frac{\ell_\nu(x_0 + a, y_0 + b)}{\ell_\nu(x_0, y_0)} = 1 + \frac{(2\alpha^2 - 1)B + 2\alpha\beta A + \alpha^2(A^2 + B^2)}{1 + B},$$

where $\alpha^2 + \beta^2 = 1$ depend on ν . Coefficientwise, therefore, $\ell/\ell_0 - 1 \ll \hat{\varepsilon} := (A + B + A^2 + B^2)/(1 - B)$, and every triangle term t satisfies $t/t_0 \ll (1 - \hat{\varepsilon})^{-3s}$, uniformly in the lattice vectors. Consequently all Taylor coefficients of T_V at τ_0 are bounded by $T_V(\tau_0)$ times the coefficients $\hat{G}_{ij}$ of $(1 - \hat{\varepsilon})^{-3s}$. This yields explicit Taylor remainders and explicit bounds for the terms omitted by a finite truncation.

Proof structure. Let $\tau_0 \in \{i, \rho\}$ be the candidate. (i) For $y > Y_0$ the collinear triples along the vector 1 give $T_V(\tau) \geq y^{3s} \sum_{j \neq k} (|j||k||j - k|)^{-2s} > T_V(\tau_0)$. (ii) Local lemma: the reflections fixing τ_0 ($x \mapsto -x$ and $\tau \mapsto 1/\bar{\tau}$ at i ; $x \mapsto 1 - x$ and the order-three rotation at ρ) force $c_{10} = c_{01} = c_{11} = c_{30} = c_{12} = 0$ in the Taylor expansion. With rigorous enclosures of $c_{20}, c_{02}, c_{21}, c_{03}$ and the majorant remainder, $T(\tau_0 + \delta) - T(\tau_0) \geq A^2(c_{20} - |c_{21}|r - \rho) + B^2(c_{02} - |c_{03}|r - \rho) > 0$ for $0 < \max(|A|, |B|) \leq r_{\text{loc}}$, where $\rho = T \sum_{i+j \geq 4} \hat{G}_{ij} r^{i+j-2}$. (iii) The rest of $[0, \frac{1}{2}] \times [0.8, Y_0]$ is covered by boxes on which the second-order Taylor polynomial at the centre, with rigorous coefficient enclosures, minus the majorant remainder exceeds an upper bound for $T(\tau_0)$. All coefficients are computed from cube-truncated sums ($R = 40$) by direct, non-FFT convolution, with an explicit tail bound and a rounding margin. Table 1 lists the data.

The crossing statement uses the same tail bound. The cube sums at $R = 100$ are evaluated by direct convolution and give $T_V(\mathbb{Z}^2) - T_V(\Lambda_\rho) \in [2.36, 2.77] \cdot 10^{-7}$ at $\nu = 3.918364$ and $\in [-3.41, -3.00] \cdot 10^{-7}$ at $\nu = 3.918366$.

Remark 1 (status of the computer-assisted parts). The local lemma of Sect. 4 uses rigorous interval arithmetic. The

Table 1 Data of the computer-assisted proofs of Theorem 4.

ν	point	$T_\nu(\tau_0)$ (enclosure)	$c_{20} \geq$	r_{loc}	boxes
3.5	ρ	[9.77234289489, 9.77240888476]	2.1286	0.0072	363
4	i	[7.58266977207, 7.58267002471]	0.4369	0.0031	531
5	i	[4.83802684613, 4.83802684730]	2.5822	0.0070	303
6	i	[3.24997431212, 3.24997431277]	4.2719	0.0076	307
7	i	[2.23344617456, 2.23344617501]	5.3445	0.0050	323
8	i	[1.55240121242, 1.55240121273]	5.8308	0.0026	385

branch-and-bound parts use IEEE double precision with explicit safety margins: 10^{-9} relative on bounds, 10^{-12} absolute on determinants, and 10^{-10} relative on sums. These exceed the rounding errors of the few operations per bound, and of sequential sums of at most $2 \cdot 10^4$ positive terms, by several orders of magnitude. As sanity checks, the three-dimensional box bounds were compared with brute-force values at 1200 random points without a single violation, and the prover fails, as it must, when asked to certify $Q < 1.49$. A re-run in full interval arithmetic is straightforward.

The same method did not succeed for the hexagonal lattice at $\nu = 3$. Close to $\nu = d$ the truncation tail decays only like R^{-4} , and at $R = 40$ the error exceeds the energy gap next to ρ .

6 Numerical phase diagram

The numerical results use an independent reference summation, described in Appendix A. It agrees with the library of [13] to better than $2 \cdot 10^{-10}$ relative in 55 test cases and reproduces the published values of [11] at $\nu = 3$.

Two dimensions. A global search over the fundamental domain gives the following picture (Figs. 2–4).

- For $4/3 < \nu < \nu^*$ the minimiser is hexagonal. For $1.4 \leq \nu \leq 1.9$ the hexagonal lattice is the best candidate with a stability of 10^{-8} across cut-off sets, using an extrapolation ansatz with the two exponent families $R^{d-2\nu}$ and $R^{2d-3\nu}$.
- At $\nu^* = 3.9183649026$ there is a first-order transition to the square lattice. Both lattices are local minima for $3.80632 < \nu < 4.27848$ (errors below $6 \cdot 10^{-5}$), and along the arc $|\tau| = 1$ the barrier is 0.14%.
- At $\nu_2 = 8.6063 \pm 0.0005$ the square lattice becomes unstable towards rectangular deformations, and the optimal aspect ratio then grows continuously towards y_∞ .

Error bars are the spread over three finite-difference steps and two cut-off sets.

Conjecture 1. In $d = 2$ the minimiser of T_ν is hexagonal for $4/3 < \nu < \nu^*$, square for $\nu^* < \nu < \nu_2$, and rectangular with aspect ratio increasing from 1 to y_∞ for $\nu > \nu_2$.

Unequal exponents. Replacing the exponent of the side $|x - y|$ by μ gives, up to normalisation, $C_{\nu/2, \nu/2, \mu/2}$. The curve of Fig. 1(b) then becomes a phase map (Fig. 5). The square phase occupies a broad band around the diagonal; for $4 \leq \nu \leq 5.5$ the hexagonal lattice returns at large μ ; the rectangular phase appears only for $\nu \gtrsim 9$ in a window $8.5 \lesssim \mu \lesssim 11$. No rhombic or oblique minimiser was found.

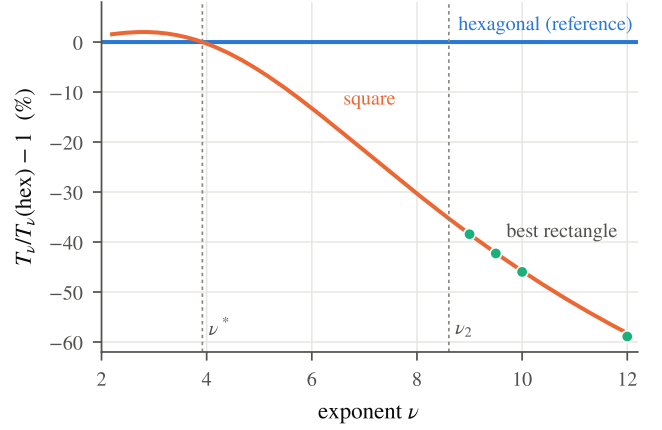


Fig. 2 Relative energy of the square lattice (line) and of the best rectangular lattice (dots) with respect to the hexagonal lattice. The zeros mark the two transitions.

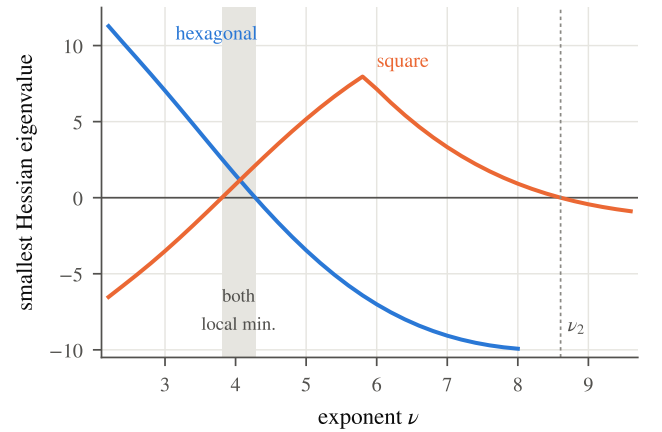


Fig. 3 Smallest Hessian eigenvalue of T_ν at the hexagonal and at the square point. Grey: both are local minima (first-order transition). Right: loss of stability of the square lattice at ν_2 .

Three dimensions. A global search over all Bravais lattices used 67 starts per exponent (FCC, BCC, SC and 64 random), Nelder–Mead and accurate re-evaluation. It returned BCC as the best lattice, to 10^{-9} relative, for every tested $\nu \in \{3.2, 3.5, 4, 4.5, 5, 6, 8, 12\}$; between 87% and 100% of the starts converged to BCC. For $2.2 \leq \nu \leq 2.8$ the ordering BCC < FCC < SC persists. The curvature of T_ν at FCC along the Bain path changes sign at $\nu_c = 3.7521 \pm 0.0001$, so FCC is a saddle point for larger ν . Optimised over $c/a \in [1.45, 1.85]$, hcp lies slightly above FCC for all tested ν ; at $\nu = 4.5$, for example, the values are 17.9724 against 17.9624, while BCC gives 17.7406. BCC is stationary along the Bain path for every exponent [11].

7 Mixed pair and three-body energies

To connect with physically motivated energies we consider $E_\lambda = Z_\nu + \lambda T_\nu$ at unit density. Here $Z_\nu = \sum' |x|^{-\nu}$ is the Epstein (Riesz) pair energy, evaluated exactly with [14], and $\lambda = c_3/c_2$. In $d = 3$, among FCC, BCC, hcp(c/a) and the Bain family, the minimiser jumps directly from FCC to BCC at $\lambda_c(\nu) \approx 0.049, 0.065, 0.103, 0.289$ for $\nu = 3.5, 4.5, 6, 9$; hcp is never optimal. In $d = 2$ the minimiser jumps from

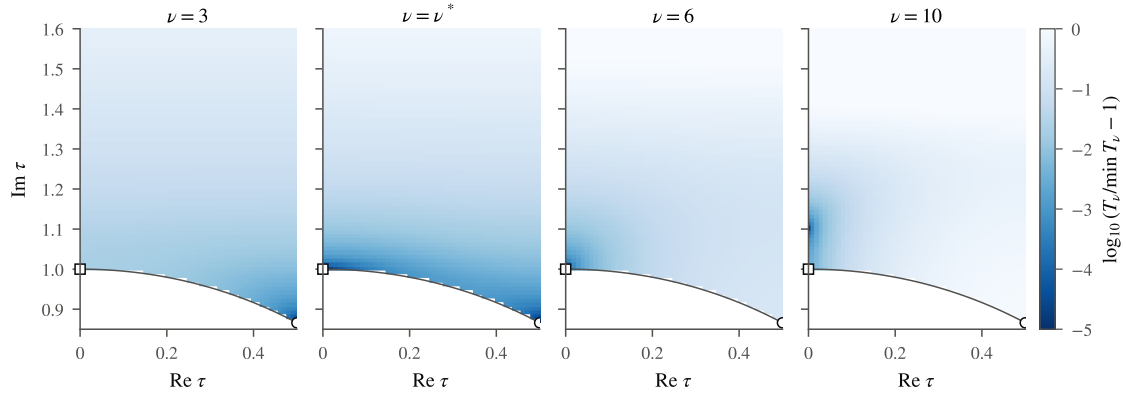


Fig. 4 Energy landscape $\log_{10}(T_\nu/\min T_\nu - 1)$ on the fundamental domain for four exponents. Circle: hexagonal point; square: square point. At $\nu = 10$ the minimum lies on the imaginary axis above $\tau = i$.

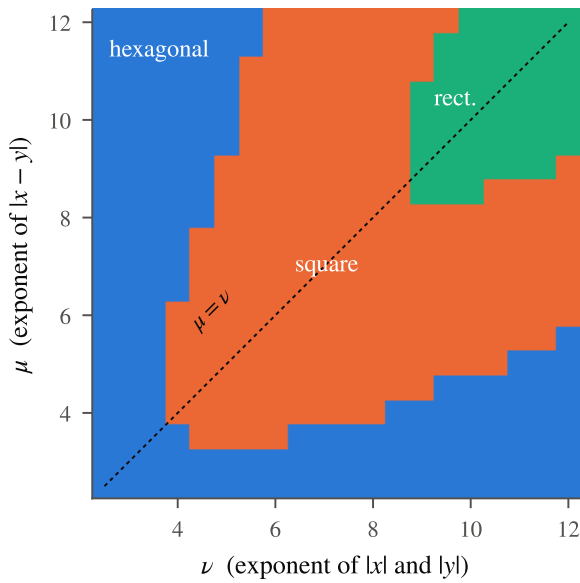


Fig. 5 Minimiser for the exponents (ν, ν, μ) on a grid of spacing 0.5. Dashed: equal exponents.

hexagonal to square at $\lambda_c \approx 1.86, 1.04, 1.51$ for $\nu = 4.5, 6, 9$, with a further transition to a rectangular lattice at very large λ for $\nu = 9$. For $\nu < \nu^*$ both terms favour the hexagonal lattice (Fig. 6).

8 Discussion

A purely three-body repulsion of product type selects different lattices than pair repulsion does. In the plane it penalises the equilateral triangles of the hexagonal lattice once the decay is steep enough. In space it favours BCC, whose smallest triangles are isosceles with one second-neighbour side, over the close-packed FCC and hcp. The steep-decay limits reduce to max-min problems for triangles. The planar one has the closed-form answer of Theorem 1; the spatial one we solved by a computer-assisted argument. Through (3) the planar results translate into statements on the minima of the modular graph functions $C_{s,s,s}$.

Several questions remain open. (i) A proof of the full planar phase diagram, in particular hexagonal optimality for all $\nu < \nu^*$, would require the finite- ν method on continua

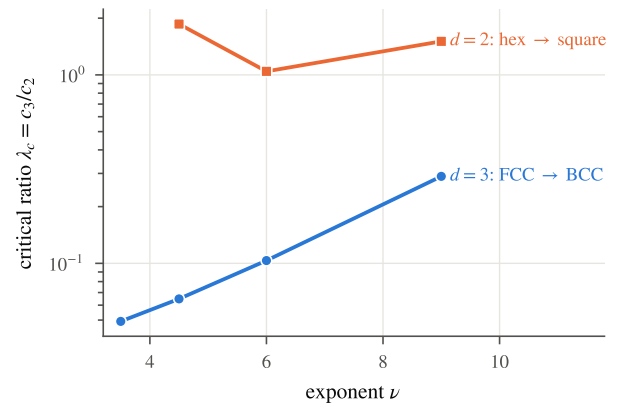


Fig. 6 Critical ratio $\lambda_c = c_3/c_2$ of the mixed energy $Z_\nu + \lambda T_\nu$.

of exponents and sharper tail bounds near $\nu = d$. (ii) An analytic proof of Theorem 3 is missing. (iii) Physically relevant three-body energies include the angular ATM factor; the mixed energies of Sect. 7 are only a first step. (iv) The rectangular phase suggests that, for $\nu > \nu_2$, the minimiser lies on the imaginary axis for all larger ν ; this is supported numerically but unproven.

Declarations

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Data availability All scripts, raw data and logs, including the branch-and-bound provers and their test suites, are available from the author on request.

Competing interests The author declares no competing interests.

A Numerical methods and error control

Reference summation. T_ν is summed over all integer labels in the cube $[-R, R]^d$ for the two free vertices, with the kernel of the third side on $[-2R, 2R]^d$. The inner convolution is evaluated by FFT for exploratory computations and by direct

convolution in the rigorous parts. The truncation error is dominated by configurations with one distant vertex and decays like R^{d-2v} for $v > d$. For $2d/3 < v \leq d$ it decays like R^{2d-3v} (with logarithms at $v = d$); Richardson extrapolation uses the corresponding exponents.

Tail bound. A term is missing only if a free vertex x lies outside the cube. For fixed x , $\sum_y |y|^{-v} |x - y|^{-v} \leq 2(|x|/2)^{-v} Z_v$, and $|Am| \geq \sigma_{\min}(A)|m|_{\infty}$, whence $0 \leq T - T_R \leq 2^{v+2} Z_v 8 \sigma_{\min}^{-2v} R^{2-2v} / (2v - 2)$ in $d = 2$.

Non-Bravais structures. For a crystal $\bigcup_j (L + s_j)$ the per-atom energy decomposes into sums $\sum_m F_{ij}(m) (F_{ik} * G_{jk})(m)$ over the Bravais lattice L . The implementation reproduces the FCC value with a four-atom cubic basis to 10^{-10} .

Validation. Besides the agreement with [13, 11] mentioned above, the reference summation reproduces the identities $C_{1,1,1} = E_3 + \zeta(3)$ (with logarithmic extrapolation, since all exponents equal d) and $C_{2,2,1} = \frac{2}{5} E_5 + \frac{\zeta(5)}{30}$ [15] to 10^{-14} and 10^{-15} .

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